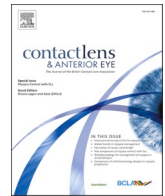




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Structural changes associated to orthokeratology: A systematic review

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ABSTRACT

Purpose: To summarize the peer-reviewed literature on corneal structural changes after orthokeratology (OK) and to analyze the quality of the studies published.**Methods:** An exhaustive search was carried out in the databases Pubmed MEDLINE, Web of Science and Scopus. Original studies in English, with a cohort or experimental design and analyzing the cellular and structural changes of the cornea after OK fitting for myopia correction were selected. The Newcastle-Ottawa Assessment Scale (NOS) tool was used to analyze the quality of the studies selected.**Results:** The search provided a total of 1837 articles, of which 12 were selected following the inclusion and exclusion criteria. All studies had good quality according to the NOS tool (mean 7.58 ± 1.31). The mean $\pm$ standard deviation follow-up period of lens wear in the studies evaluated was 19.27 ± 33.97 months. The following reversible changes after OK have been reported: reduction in central epithelial basal cells, an increase in height and especially in width of superficial epithelial cells, as well as an increase in central stromal thickness and in the number of active keratocytes. Concerning corneal endothelium, only changes in terms of polymegathism were reported that were compatible with those observed for other types of contact lens wear. In relation to corneal sensitivity, there was a reduction during the use of OK, as well as in the nervous density of the sub-basal plexus at the central level, with some contradictory outcomes concerning the reversibility of these changes.**Conclusion:** OK produces reversible structural changes in the central epithelium and central anterior corneal stroma, as well as a decrease in the nerve density of the sub-basal plexus and corneal sensitivity. The quality of published studies evaluating these aspects is good, although more studies are needed to evaluate longer follow-up changes.

1. Introduction

Orthokeratology (OK) is a refractive treatment aiming at correcting the refractive errors by a short-term modification of the corneal curvature. [1] This term was first used in 1962 by Newton Wesley at the congress of the International Society of Contact Lens Specialists held in Chicago. Afterwards, Kerns modified the definition of OK as follows: "Orthokeratology is the reduction, modification or elimination of refractive errors through the programmed application of contact lenses" [2]. In 1962, George Jessen was the first to use rigid PMMA contact lenses to reduce the refractive error by flattening (or steepening) the base radius of the lens by increments of 0.05 mm (by 0.25 D), until

achieving the total refractive correction required by the patient [3]. However, OK only became popular at the beginning of the 21 st century due to the availability of new materials with higher oxygen permeability, improvements in geometric designs, and the certification of an OK contact lens design by the American FDA in 2002. [4,]⁵

The predominant hypothesis to explain the refractive correction achieved with OK for myopia is that hydrodynamic forces generated by the presence of the tear film between the cornea and the lens generates suction and flattening forces on the corneal tissue. According to this theory, areas in which the tear layer is between 5 and 10 microns in thickness will promote corneal flattening, while areas in which the cornea presents an accumulation of tear under the lens will steep the

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cornea. [2] These differences in tear film thickness below the lens are necessary to generate this corneal molding.

In 1998, Swarbrick et al. proposed another mechanism explaining the OK effect, consisting on a redistribution of superficial corneal tissue instead of a purely mechanical bending of the cornea, as the central epithelium thinned while the anterior epithelium and stroma in the paracentral zone thickened. [5] However, all corneal structural and histological changes occurring with OK are not fully understood by clinicians, which is crucial to minimize and manage potential complications and side effects [4].

There is a necessity of discerning and compiling the information about the specific corneal structural changes occurring after OK in terms of changes in corneal layers, epithelial and endothelial distribution, corneal nerves or collagen fiber disposition to obtain a more consistent knowledge on the mechanism of action of this contact lens technique of refractive correction. The objective of this review was to summarize the different corneal structural and histological changes that occur at the corneal level after overnight OK and to analyze the quality of the studies published in this regard using a validated instrument for such purpose.

2. Methods

A systematic literature search has been conducted on the following databases: Pubmed MEDLINE, Web of Science (WOS) and Scopus (all databases). The search was limited to studies published from 2010 to 2020, focusing on the last 10 years. The Boolean search criterion was: ("orthokeratology" OR "corneal refractive therapy") NOT/AND NOT ("atropine" OR "myopia control" OR "myopia progression").

The inclusion criteria were: i) original articles whose main objective was to report changes in the corneal structure after OK; ii) non-randomized experimental studies with sample populations including all ages from children to adults. Studies whose main objective was not to evaluate structural changes in the cornea after OK were excluded. Likewise, studies about expert guides or opinions, non-original articles and studies in a language different than English were excluded. Although the scope of the bibliographic search was broad, this strategy was chosen to avoid documentary silence and make sure that all relevant publications were included.

The selected studies were qualitatively analyzed using the Newcastle-Ottawa Assessment Scale (NOS) tool for cohort studies. [6,7] The NOS, which is an ongoing collaboration between the Universities of Newcastle, Australia and Ottawa, is one of two tools recommended by the Cochrane Collaboration to evaluate the methodological quality of non-randomized studies [8]. The studies were evaluated in three main areas: participant selection, comparability, and outcome. A study can be awarded a maximum of one star for each numbered item within the selection and outcome categories, and a maximum of two stars for comparability. The thresholds for converting the NOS scale to the Agency for Healthcare Research and Quality (AHRQ) standards (good quality, fair, or poor) are shown in Table 1.

3. Results

The search was performed on December 7th, 2019. For the elaboration of the flow chart and the data collection process, the guidelines of

Table 1
NOS tool quality scoring guide and conversion to the AHRQ standards.

Quality	Item Score
Good	3 or 4 stars in Selection / AND / 1 or 2 stars in the Comparability / AND / 2 or 3 stars in Results
Fair	2 stars in Selection / AND / 1 or 2 stars in Comparability / AND / 2 or 3 stars in Results
Poor	0 or 1 star in Selection / OR / 0 stars in Comparability / OR / 0 or 1 star in Results

the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) declaration were followed. [9] Fig. 1 shows the flow chart followed during the search.

Using the Boolean search criterion, a total of 1837 articles were found. After removing duplicates, 429 studies remained. Following the inclusion criteria, 29 articles were initially selected for a more comprehensive study and to confirm whether they should be included or not. Of the 29 articles, 21 did not fulfill the inclusion criteria and were excluded from further analysis. Eventually, 4 studies were included after a manual search evaluating the list references of the studies previously retrieved in the review. The final number of included articles to be analyzed qualitatively was 12. [11–22]

In relation to the design of selected studies, all of them were prospective cohort studies, except one in which the information about the treatment with OK was collected retrospectively. [11] The results in terms of quality of the studies evaluated with the NOS tool are outlined in Table 3. The average score of all studies was 7.58 ± 1.31 . All of them had good quality according to the NOS score thresholds. The maximum score was 9 and it was obtained for 4 studies. [13,16,19,21] The minimum scores were 5 and 6 for two studies of the same author, due to its methodology and lack of information provided. [11,]17

The average time of OK treatment for all studies was 19.27 ± 33.97 months, with a minimum of 1 night of use and a maximum of a 9.9 ± 4.4 years of use. The main measuring instruments used were the confocal microscope for studies that measured the structural changes at the cellular level, and the Cochet-Bonnet (COBO) esthesiometer for those evaluating changes in corneal sensitivity.

The sample size, clinical follow-up after fitting, the fitted lens (manufacturer, design and technical parameters), the main structural changes reported, the measuring instruments, and the main outcomes of each of the studies reviewed are summarized in Table 2. Of these 12 studies, 3 of them evaluated changes in the density and number of corneal cells, as well as the number of keratocytes activated in corneal stroma. [13,15,19] One article evaluated the changes and the cellular integrity in the corneal endothelium [20]. The remaining 8 articles assessed changes on corneal sensitivity and sub-basal nerve plexus. [10–12], [14,16–18] [21], The OK lenses used in each study were Boston XO and XO2 (8/12 studies), CRT (4/12 studies) and Boston EM (1/12 studies).

All the studies agree that, after overnight OK, there is a significant change in the epithelial basal cells, with an increase in height and

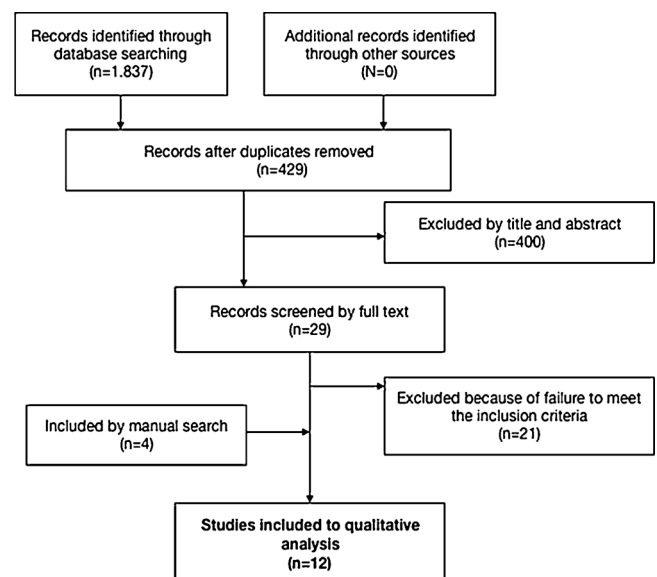


Fig. 1. Flow chart (PRISMA) for the selection process of the relevant articles that were included in the systematic review for further quality analysis.

Table 2

Main characteristics of 12 studies included. OK, orthokeratology; DK, oxygen permeability; GP, conventional rigid gas-permeable lens; COBO, Cochet-Bonnet aesthesiometer; NCCA, non-contact corneal aesthesiometer; NFD, nerve fiber density; GNFO, Global nerve fiber orientation; SBNP, sub-basal nerve plexus.

Study	SAMPLE (N)	Age (years)	Follow-up	Lens fitted	Structural Changes Evaluated	Measuring instruments	Main outcomes
Hiraoka T et al. (Cornea, 2004)	52 eyes	16.9 ± 8.5	1 year	Emerald (Boston Equalens II) (DK = 85)	i Endothelial cell density. ii cell area iii Percentage of hexagonal cells.	Specular microscopy	None of the parameters changed significantly after one year with orthokeratology, so this treatment did not affect the density or morphology of corneal endothelial cells.
Hiraoka T et al. (Cornea, 2009)	34 eyes	23.5 ± 3.2	3 months	Boston XO2 (DK = 141)	Corneal sensation in 5 different corneal locations: one central point and 4 peripheral points at the superior, inferior, temporal, and nasal positions 2 mm away from the limbus	Cochet-Bonnet esthesiometer	Overnight orthokeratology significantly reduced corneal sensation in every location evaluated. There was no significant correlation between the loss of corneal sensation and the amount of myopic correction and patient age.
Zhong X et al. (Cornea, 2009)	–5-year lens wear (n = 26, 51 eyes) –1-night lens wear (n = 30, 60 eyes)	19.4 ± 5.0	–5-year lens wear – 1-night lens wear	Boston XO (DK = 145)	Epithelial and stromal thickness, basal cell density, keratocyte density in anterior and posterior stroma were measured. Density of endothelial cells.	Confocal microscopy	- After an 8-h lens wear, the cell morphology, thickness, and basal cell density in central epithelium and endothelium did not significantly change. The density of keratocytes decreased in the anterior and posterior stroma (p < 0.001). - After 5 years, the density of the basal cells and the number of keratocytes of the epithelium decreased significantly (p < 0.001). The keratocyte morphology and stromal thickness in the central cornea, the Bowman layer and the endothelium (morphology, cell density, and proportion of the hexagon cells, p < 0.15) did not change.
Nieto-Bona A et al. (CORNEA, 2011)	- OK group (n = 27 eyes). - Control group (n = 10 eyes)	25.4 ± 2.6	15 days and 1 month	CRT (DK = 100)	Cell density, percentages of wing, basal and superficial cells from the corneal epithelium, stroma (anterior, middle and posterior), the Bowman layer, the subbasal nerves plexus and cell counts and to assess polymegathism and pleomorphism in the endothelium.	Confocal Microscopy	No significant changes in either endothelial cell or stromal cell density were observed after 1 month of OK. Basal epithelial cells were significantly reduced (p < 0.01), and epithelial wing and superficial cells showed enhanced visibility (p < 0.05). Superficial cells increased in height and width, the width increase after 1 month being significant (p < 0.01). Increase of activated keratocyte number after 15 days and 1 month. Decrease of nerve fiber number after 1 month with OK.
Nieto-Bona A et al. (Curr Eye Res, 2011)	15 eyes	25.5 ± 2.5	15 days, 1 month, 3 months, 6 months and 1 year. 1 month after interruption OK wear.	CRT (DK = 100)	Epithelium basal density, cell widths and heights (basal and superficial). Keratocyte activation. Nerve fiber number. Stromal keratocyte density (anterior, middle, and posterior). Endothelium cell density, polymegathism and pleomorphism.	Confocal microscopy	Basal epithelium cell densities significantly fell (p < 0.01). Superficial cells increased in height and width being significant after 1 year of orthokeratology (p < 0.01). All epithelial cell changes returned to baseline. Stromal activated keratocytes increased but baseline values were recovered. No significant changes in endothelial cell density but polymegathism increased significantly and baseline

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Table 2 (continued)

Study	SAMPLE (N)	Age (years)	Follow-up	Lens fitted	Structural Changes Evaluated	Measuring instruments	Main outcomes
Lum E et al. (Optom Vis Sci, 2013)	20 subjects	Mean age, 24 years; range, 19–39 years	1 night (~8 h)	OK BE (Boston XO, DK = 45) J&J Acuvue Advance (DK = 50) GP Capricornia J-Contour (Boston XO, DK = 46)	Changes in central corneal sensitivity with three different lens type wear.	COBO aesthesiometer and NCCA	values were not recovered ($p < 0.01$). There was a significant increase in threshold from baseline only for the OK lens type ($p = 0.006$) with COBO. There was no correlation between the change from baseline in corneal sensation threshold and corneal refractive power, asphericity following OK lens wear. NCCA didn't show a significant difference in corneal sensitivity between lens type, and didn't show comparable outcomes with COBO.
Lum E et al. (Ocul Surf, 2017, A)	33 subjects (OK group, $n = 16$; GP group, $n = 8$; NL group, $n = 9$)	26 ± 9	3 months	BE (Boston XO2, DK = 141)	Nerve fiber number. Corneal sensitivity and NFD at the corneal apex and 2.5 mm temporal to the apex. GNFO	In vivo Corneal Confocal Microscopy. COBO aesthesiometer.	Significant changes in NFD, GNFO and corneal sensitivity were observed during OK lens wear over the study period. In the central cornea, both NFD and corneal sensitivity decreased. GNFO was rotated in a clockwise during the OK lens wear. In GP lens users, the corneal sensitivity decreased.
Lum E et al. (Ocul Surf, 2017, B)	16 subjects	28 ± 9	3 months after non-lens wear	BE (Boston XO2, DK = 141)	Nerve fiber number. Corneal sensitivity and NFD at the corneal apex and 2.5 mm temporal to the apex. GNFO	In vivo Corneal Confocal Microscopy. COBO aesthesiometer.	NFD and corneal sensitivity increased during the 3-month non-lens wear period in the central cornea ($p < 0.01$), with no significant changes in the mid-peripheral cornea. GNFO did not change significantly at the central cornea during the 3-month non-lens wear period, but altered in a counter-clockwise rotational direction at the mid-peripheral cornea ($p < 0.05$).
Lum E et al. (Eye Contact Lens, 2017)	54 subjects	53 ± 9	Lens Wear Experience (retrospectively): SCL 9.9 ± 4.4 years OK 4.2 ± 3.1 years	Not Reported	Three measure groups: nonlens, soft lens, and OK lens wearers. Corneal sensitivity at the corneal apex and 2.5 mm temporal to the apex. Nerve morphology at the corneal apex and 2.5 mm temporal to the apex, and sub-basal NFD.	In Vivo Corneal Confocal Microscopy. COBO aesthesiometer	Central threshold was significantly higher in the OK than NL group ($p = 0.048$). Midperipheral threshold was not different between the three groups ($P.0.05$). Central NFD was significantly less in the OK than nonlens and soft lens groups ($p < 0.001$). Sensitivity and NFD were reduced centrally compared with the mid-periphery.
Jin W et al. (J Ophtalmol, 2018)	30 subjects	11.07 ± 2.13	3 months	Alpha (Boston EM, DK = 104)	Corneal sensitivity at the corneal apex and 2.5 mm temporal to the apex.	COBO aesthesiometer	Central corneal sensitivity decrease suffered a significantly within the first month of the OK treatment period but returned to the baseline level at 3 months ($p = 0.039$), while no statistically significant difference occurred in temporal sensitivity. The baseline of central corneal sensitivity correlated with age ($r = -0.369$, $P = 0.045$).
Nombela-Palomo M et al. (Eye Contact Lens, 2018, A)	- CRT $n = 35$. - SeeFree $n = 21$. - Control group $n = 15$	23.9 ± 3.7	1 month	- CRT (HDS, DK = 100) - SeeFree (Boston XO2, DK = 141)	Corneal sensitivity at the corneal apex and 3,2 mm temporal to the apex. SBNP thickness. Reflectivity Global and Local. Nerve number, reflectivity and density. Length.	In Vivo Corneal Confocal Microscopy. COBO aesthesiometer	Significant reductions in nerves number in the central cornea in the groups CRT ($p = 0.029$) and SeeFree ($p = 0.043$) and in central corneal sensitivity in CRT ($P = 0.047$) along with significant

(continued on next page)

Table 2 (continued)

Study	SAMPLE (N)	Age (years)	Follow-up	Lens fitted	Structural Changes Evaluated	Measuring instruments	Main outcomes
					Orientation. Langerhans density. Width. Tortuosity objective and subjective.		increases in central and midperipheral corneal Langerhans cell counts in SeeFree (P = 0.001 and 0.048, respectively).
Nombela-Palomo M et al. (Eye Contact Lens, 2018 B)	- CRT n = 17. - SeeFree n = 15. - Control group n=15	25.1 ± 3.7	1 year of treatment. 1 month OK interruption	- CRT (HDS, DK = 100) - SeeFree (Boston XO2, DK = 141)	Corneal sensitivity at the corneal apex and 3,2 mm temporal to the apex. SBNP thickness. Reflectivity Global and Local. Nerve number, reflectivity and density. Length. Orientation. Langerhans density. Width. Tortuosity objective and subjective.	In Vivo Corneal Confocal Microscopy. COBO aesthesiometer	Over the year lens wear, significant reductions in the central and mid-peripheral cornea, respectively, in SBNP nerve density (p = 0.001 and p = 0.006) and number of nerves (p < 0.001 and p = 0.001) and differences central objective tortuosity (p = 0.002) were detected. After lens removal, baseline values of NFD (p = 0.024 and p = 0.001) and number of nerves (p = 0.021 and P < 0.001) for the central and mid-peripheral cornea, respectively, were not recovered. At one month post-treatment, a difference was observed in central corneal sensitivity (p = 0.045) and mid-peripheral Langerhans cell density (p = 0.033), and from baseline in mid-peripheral objective tortuosity (p = 0.049). Direct correlation was detected at one year between nerve density and tortuosity both in the central (p < 0.01; r = 0.69) and midperipheral cornea (P < 0.01; r = 0.76).

Table 3

Quality outcomes with NOS tool.

STUDY	SELECTION	COMPARABILITY	OUTCOME	QUALITY
Hiraoka T et al. (Cornea, 2004)	☆☆☆	☆	☆☆☆	7
Hiraoka T et al. (Cornea, 2009)	☆☆☆	☆☆	☆☆☆	8
Zhong X et al. (Cornea, 2009)	☆☆☆	☆☆	☆☆	7
Nieto-Bona A et al. (Cornea, 2011)	☆☆☆☆	☆☆	☆☆☆	9
Nieto-Bona A et al. (Curr Eye Res, 2011)	☆☆☆☆	☆☆	☆☆☆	9
Lum E et al. (Optom Vis Sci, 2013)	☆☆☆	☆	☆☆	6
Lum E et al. (Ocul Surf, 2017, A)	☆☆☆☆	☆	☆☆☆	8
Lum E et al. (Ocul Surf, 2017, B)	☆☆☆	☆	☆☆☆	7
Lum E et al. (Eye Contact Lens, 2017)	☆	☆	☆☆☆	5
Jin W et al. (J Ophtalmol, 2018)	☆☆☆	☆	☆☆☆	7
Nombela-Palomo M et al. (Eye Contact Lens, 2018, A)	☆☆☆☆	☆☆	☆☆☆	9
Nombela-Palomo M et al. (Eye Contact Lens, 2018 B)	☆☆☆☆	☆☆	☆☆☆	9

especially in width, an increase in stromal thickness and in the number of active keratocytes. Cellular polymegathism] increases in corneal endothelium, which even remains after the cease of the use of OK lenses. [13,20] This endothelial change is within the range reported by other authors for the conventional use of gas-permeable contact lenses. [22,23] Corneal sensitivity and nervous density of the sub-basal plexus at the central cornea are also reduced with the use of OK. However, the nervous density of the sub-basal in the mid-peripheral cornea does not present significant changes [10,11,16].

4. Discussion

The purpose of the current systematic review was to summarize the available relevant evidence related to the main structural corneal changes occurring after OK and to evaluate its quality using a validated tool. The comprehensive analysis of this peer-reviewed literature confirms that the main structural corneal changes occurring after OK are generated in the epithelium and anterior stroma, which are the regions of the cornea experiencing the most relevant geometrical deformation during OK. These structural changes are predominantly associated with

a central and paracentral decrease of corneal thickness. [1,2,5,24,25] Concerning corneal endothelium, no clinically significant structural changes have been reported, with only an increase in cell polymegathism during the OK treatment. [13,19] Similar levels of pleomorphism and endothelial polymegathism have already been reported by other authors in long-term contact lens wearers, especially of rigid gas-permeable contact lenses, that improve after cessation of contact lens wear, although not reaching the baseline values. [22,]23

Regarding the tissue distribution changes, Nieto-Bona et al. showed a significant reduction of middle stroma keratocyte density after 15 days of OK lens wear, with non-significant reduction also in anterior and posterior stroma keratocyte density. [19] Likewise, these authors also reported a reduction of epithelial cell density, but basal cell heights and widths did not change significantly during the first month of OK lens wear [19]. However, an increase in the irregularity of the basal cell network distribution and cell nuclear size was observed [19]. In contrast, these same authors described an increase in heights and widths of superficial epithelial cells. Visibility percentages of folds, microdots and Langerhans cells were also reported to peak at 15 days but returning to baseline after 1 month of OK [19]. All these changes were associated to a progressive reduction of central epithelial and Bowman layer thicknesses as well as an increase in mid-peripheral epithelial and central stromal thickness. [13,]19 The authors attributed the epithelial changes to cell redistribution, suggesting that the corneal epithelium was the principal structure affected by the mechanical forces exerted by the OK lenses. These epithelial changes have been reported by Zhong et al. in an illustrated way using images from confocal microscopy.¹⁵ All epithelial cell changes returned to baseline after OK lens discontinuation, although epithelial thickness baseline values were not fully recovered. [13]

No studies are available comparing the changes in the corneal structure or at histological level using OK lenses and other types of contact lenses and therefore consistent conclusions concerning this issue cannot be extracted. However, there are studies comparing changes in corneal sensitivity with OK lenses and with conventional soft and rigid gas permeable lenses. This comparison of OK with conventional lenses revealed the presence of a higher sensitivity threshold in the central cornea with OK lenses compared to conventional contact lenses. At the same time, a lower nerve fiber density was observed in the central cornea with OK lenses compared to soft or gas permeable lenses. These observations indicate that that wearing OK lenses induces a decrease of corneal sensitivity, [11,17] which was found to plateau by day 30 after initiating the OK treatment [18]. A reduced nerve fiber density was associated with this reduced corneal sensitivity [18]. Furthermore, global nerve fiber orientation also experienced changes with OK, rotating in a clockwise direction at day 30 of treatment in the mid-peripheral cornea, with further rotation at day 90 [18]. Remarkably, Nombela-Plomo et al. (2018) showed that after one year of use of OK lenses, the decrease in corneal sensitivity induced by the refractive treatment was maintained even one month after lens discontinuation. [16] However, other authors reported a recovery of corneal sensitivity at 3 months after the OK lens discontinuation, observing a stabilization of the corneal sensitivity after long-term use of OK, explained by a neural adaptation to the effect exerted by the lens on corneal epithelium [18–29].

Other changes that have been reported in some case reports by some authors is the presence of an iron ring in the midperipheral corneal epithelium of some patients undergoing OK. [26,]27 This rare condition deserves to be considered and studied further although it does seem to be related to any risk.

Although qualitative analysis of these studies outlined that 10 out of 12 had a good methodological quality and, therefore, that their results can be extrapolated to the population, some considerations must be accounted for. There exist only 4 published studies evaluating the structural changes on the cornea after OK, and only 8 studies evaluating changes on corneal sensitivity. The prospective evaluation periods

ranged from few months to one year, except for a single study in which the treatment was maintained for 5 years. Although the longest follow-up period was 9 years, it is unknown whether the effects of the treatment and its rigorous compliance were correctly controlled in a retrospective way. [11]

Reversibility of the changes induced with OK were only evaluated in 2 of the 12 studies analyzed. [13,]16 Both studies concluded that most part of pre-treatment baseline values are recovered. Topographic corneal values, refractive error and contrast sensitivity were found to be recovered within a period of less than 8 weeks after OK discontinuation [28]. As previously mentioned, there are some doubts about whether corneal sensitivity changes as well as changes in the number of corneal nerve fibers are totally reversible in the long term. More research is still needed on this issue as well as on the real clinical impact of these corneal nerve alterations.

In summary, orthokeratology does produce structural changes in the epithelium and anterior corneal stroma, as well as a decrease in the nerve density of the sub-basal plexus and corneal sensitivity during the OK use, according to the scientific literature. The corneal endothelium does not appear to undergo relevant structural changes, with a polymegathism of the cells within ranges described for other types of contact lens wear. Overall, and although the quality of published studies is good, more studies are needed to evaluate cell changes and corneal morphology with both longer follow-up periods and longer study periods of the reversibility of the OK effect.

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Declaration of Competing Interest

The authors have no proprietary or commercial interest in the medical devices that are involved in this manuscript.

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